End-to-End Multimodal Transformers for Multi-Cohort Alzheimer's Classification

Tanguy Vansnick
Faculty of Engineering - ILIA
University of Mons
Rue de Houdain 9, 7000 Mons, Belgium
tanguy.vansnick@umons.ac.be

Maxime Gloesener
Faculty of Engineering - ILIA
University of Mons
Rue de Houdain 9, 7000 Mons, Belgium
maxime.gloesener@umons.ac.be

Otmane Amel
Faculty of Engineering - ILIA
University of Mons
Rue de Houdain 9, 7000 Mons, Belgium
otmane.amel@umons.ac.be

Vito Tota
Department of Neuroscience
University of Mons
20 Place du Parc, 7000 Mons, Belgium
vito.tota@umons.ac.be

Mathis Delehouzee
Faculty of Engineering - ILIA
University of Mons
Rue de Houdain 9, 7000 Mons, Belgium
mathis.delehouzee@umons.ac.be

Saïd Mahmoudi
Faculty of Engineering - ILIA
University of Mons
Rue de Houdain 9, 7000 Mons, Belgium
said.mahmoudi@umons.ac.be

for the Alzheimer's Disease
Neuroimaging Initiative*

***Abstract*—Deep learning for Alzheimer's disease (AD) classification from MRI has shown promise, yet recent reviews reveal that fewer than 20% of studies employ external validation, with data leakage (e.g., same subjects in train and test sets) inflating reported accuracies to 95-99%. Studies with rigorous subject-level splitting achieve 66-90%. We present an end-to-end multimodal deep learning framework that processes both 3D MRI scans and clinical tabular data. The framework combines a Vision Transformer (ViT) for imaging analysis with a Feature Tokenizer Transformer (FT-Transformer) for clinical features, fusing both modalities through bidirectional cross-attention. This multimodal fusion enables the model to leverage complementary information from structural brain imaging and patient clinical profiles. We validate on 6,065 subjects from three independent cohorts (ADNI, OASIS, NACC-SCAN) using strict subject-level 5-fold cross-validation. Our approach achieves 92.4% accuracy (AUC: 0.96) for distinguishing cognitively normal subjects from AD-trajectory patients (including MCI patients who later developed AD), and 93.3% on established AD cases. External validation across datasets reveals that single-cohort models suffer severe performance degradation (accuracy drops of 25-43 points), while our multi-cohort approach maintains robust generalization.**

***Index Terms*—Alzheimer's disease, multimodal fusion, Vision Transformer, deep learning, MRI classification**

I. Introduction

Alzheimer's disease (AD) is the most common cause of dementia, contributing to 60-70% of the 57 million dementia cases worldwide [1]. Early and accurate diagnosis is crucial for patient care, clinical trials enrollment, and treatment planning. Early diagnosis increasingly relies on biomarkers—CSF analysis requiring lumbar puncture, or amyloid PET imaging with radioactive tracers—which are impractical for routine screening. Structural MRI combined with clinical tabular data offers a practical alternative, capturing characteristic patterns of brain atrophy alongside demographic and cognitive assessments without invasive procedures or radiation exposure.

Deep learning approaches for AD classification have proliferated in recent years, with many studies reporting accuracies exceeding 95%. However, a recent scoping review by Young et al. [2] revealed that fewer than 20% of published studies employ external validation, and methodological issues remain widespread. The primary issue is data leakage: when data from the same subject appears in both training and test sets, models learn to recognize individual brain morphology rather than disease-specific patterns.

Yagis et al. [3] quantified this effect: when 2D slices from the same subject appear in both train and test sets, models achieve 96% accuracy even with *randomly assigned labels*—demonstrating that they learn to recognize individual brain morphology rather than disease patterns. With proper subject-level splitting, random-label accuracy drops to 50%, and reported accuracies for real labels fall to 66-90% [2].

Combining multiple data modalities has shown promise for improving AD classification. Clinical tabular data (demographics, cognitive assessments) provides complementary information to structural MRI. However, diagnostic screening scores like the Mini-Mental State Examination (MMSE) are strongly correlated with cognitive status. MMSE, a 30-point clinician-administered test, showed median scores of 29 for cognitively normal (CN) subjects versus 23.5 for AD patients in the ADNI cohort [4]. We explore combining modern transformer architectures—specifically ViT for 3D MRI and FT-Transformer for tabular data—while excluding such diagnostic screening scores (MMSE, CDR) to ensure the model learns from structural imaging patterns and neuropsychological assessments rather than diagnostic proxies.

In this work, we present a fully end-to-end multimodal deep

learning framework that combines:

- Vision Transformer (ViT) with masked autoencoder (MAE) pretraining for 3D MRI feature extraction
- Feature Tokenizer Transformer (FT-Transformer) for tabular clinical data
- Cross-attention fusion for learned modality interaction

Our contributions are: (1) an end-to-end multimodal framework combining 3D ViT and FT-Transformer for AD classification; (2) validation on 6,065 subjects from three independent cohorts (ADNI, OASIS, NACC) with strict subject-level splitting; (3) cross-task generalization: training on the AD-trajectory (including MCI converters) achieves 92.4% on established AD detection; and (4) comprehensive ablation studies demonstrating the value of multimodal fusion.

II. Related Work

A. Deep Learning for AD Classification

Convolutional neural networks (CNNs) have dominated AD classification from MRI. Wen et al. [5] provided a comprehensive review, showing that 3D CNNs achieve up to 89% balanced accuracy with proper methodology. More recently, Vision Transformers (ViT) [6] have been adapted for 3D medical imaging. VECNN [7] combined ViT with CNNs, reaching 94.1% on ADNI.

B. Methodological Concerns

A critical issue in AD classification is data leakage, where test subjects' data inadvertently influences training. Yagis et al. [3] quantified this effect: improper data splitting inflates accuracy by 25-55%, with models achieving 96% accuracy even on randomly assigned labels through memorization of individual brain morphology. Young et al. [2] found that fewer than 20% of published studies employ external validation, with studies lacking proper subject-level splitting reporting 95-99% accuracy while rigorous studies achieve 66-90%.

C. Transformer Architectures

Vision Transformers (ViT) [6] divide images into patches, treating each as a token for self-attention. The self-attention mechanism allows direct modeling of long-range spatial dependencies, which may be beneficial for capturing distributed atrophy patterns characteristic of AD across multiple brain regions. Masked autoencoder (MAE) pretraining [8] enables ViT to learn robust representations from unlabeled data by reconstructing masked patches. Kunanbayev et al. [10] explored MAE pretraining for medical imaging in low-data regimes, training ViT-Base models on medical datasets (BRATS 2023, IXI, OASIS3, HCP) with masking ratios of 25%, 50%, and 75%. The 75% masking ratio achieved the highest performance gains (up to 13.6% improvement on ADNI), and the authors released pretrained weights which we use to initialize our MRI encoder.

For tabular data, the FT-Transformer [9] embeds each feature as a separate token, enabling attention across features. This demonstrates strong performance compared to traditional MLPs across diverse tabular benchmarks.

D. Multimodal Fusion

Combining imaging with clinical data improves classification performance. Qiu et al. [4] demonstrated this with an end-to-end FCN-MLP framework that achieved 96.8% accuracy on ADNI by integrating MRI features with age, sex, and MMSE scores. However, diagnostic scores like MMSE (median: 29 for CN vs 23.5 for AD) are strongly predictive of cognitive status, raising concerns about whether such models learn disease patterns versus cognitive assessment correlations. In a larger follow-up, Qiu et al. [11] scaled to 8,916 subjects across 8 cohorts using a hybrid architecture that concatenates CNN-derived imaging features with CatBoost for tabular data fusion.

We advance this line of work by combining state-of-the-art architectures for both modalities: 3D Vision Transformer with masked autoencoder pretraining for imaging and Feature Tokenizer Transformer for clinical data, integrated through bidirectional cross-attention. We exclude diagnostic scores (MMSE, CDR) to ensure the model learns from structural neuroimaging and non-cognitive clinical features rather than test-score proxies of the diagnostic label.

III. Methods

A. Datasets

We aggregated data from three publicly available cohorts:

ADNI (Alzheimer's Disease Neuroimaging Initiative) [12]: A longitudinal study collecting multimodal neuroimaging and clinical data to track Alzheimer's disease progression. We included 903 subjects (424 CN, 479 AD-trajectory).

OASIS-3 (Open Access Series of Imaging Studies) [13]: A longitudinal neuroimaging dataset from Washington University including cognitively normal older adults and individuals with cognitive decline. We included 1,030 subjects (742 CN, 288 AD-trajectory).

NACC (National Alzheimer's Coordinating Center) [14]: The largest US Alzheimer's database, aggregating data from over 40 Alzheimer's Disease Research Centers (ADRCs). MRI data were obtained from SCAN (Standardized Centralized Alzheimer's & Related Dementias Neuroimaging), NACC's centralized neuroimaging repository. We included 4,132 subjects (3,581 CN, 551 AD-trajectory).

Classification Groups: The AD-trajectory group includes both clinically diagnosed AD patients (established AD) and MCI subjects who subsequently progressed to AD (MCI converters). This grouping is grounded in the 2018 NIA-AA Research Framework [16], which defines AD as a biological continuum rather than a clinical syndrome—individuals with AD pathology are on the Alzheimer's continuum regardless of clinical stage. Longitudinal neuroimaging studies have demonstrated that MCI converters exhibit AD-like atrophy patterns detectable on structural MRI at baseline [17], [18]. The combined dataset comprises 6,065 subjects (4,747 CN, 1,318 AD-trajectory: 78.3% vs 21.7%). Table I summarizes the dataset characteristics.

TABLE I
MULTI-COHORT DATASET COMPOSITION AND SUBJECT DISTRIBUTION

Cohort	Total	CN	AD-traj	% Total
ADNI	903	424	479	14.9%
OASIS	1,030	742	288	17.0%
NACC	4,132	3,581	551	68.1%
Total	**6,065**	**4,747**	**1,318**	100%

B. Data Splitting Strategy

To prevent data leakage, we employed strict subject-level splitting with 5-fold stratified cross-validation:

- Each subject appears in exactly one split (train/val/test)
- Only the first visit per subject is used to avoid temporal leakage from repeated scans
- All 6,065 subjects are pooled and split via stratified 5-fold CV (stratified by class label; cohort distribution varies naturally across folds)
- Each fold: 80% train+val / 20% test (rotating test sets)
- Within each fold: train is split 90/10 into train/val
- Final splits per fold: 72% train / 8% val / 20% test (4,367 / 485 / 1,213 subjects)
- Experiments repeated with 3 random seeds (42, 123, 456)
- Tabular feature normalization (StandardScaler) fitted only on training split to prevent information leakage

This full cross-validation approach ensures every subject serves as a test sample exactly once per seed. The methodology follows recommendations from Wen et al. [5] and addresses concerns raised by recent systematic reviews [3].

C. MRI Preprocessing

T1-weighted MRI scans underwent standardized preprocessing (Fig. 1):

1) Conversion from DICOM to NIfTI format
2) N4 bias field correction to remove intensity inhomogeneities
3) Non-linear registration to MNI-152 template using SyN (1.75mm isotropic)
4) Skull stripping using SynthStrip [15]
5) Resampling to 128×128×128 voxels

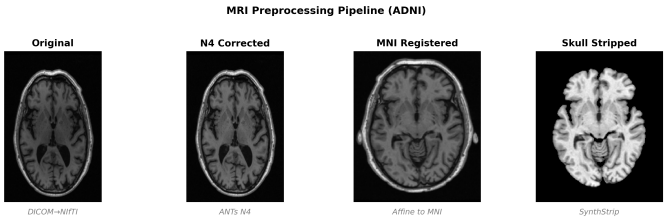

Fig. 1. MRI preprocessing pipeline showing (left to right): original T1-weighted scan, N4 bias-corrected image, MNI-152 registered volume, and skull-stripped brain.

D. Clinical Features

We harmonized and extracted 16 clinical features consistently available across all three cohorts (ADNI, OASIS, NACC):

- **Demographics (4):** Age, sex, education years, marital status
- **Neuropsychological tests (6):** Category fluency (animals), Trail Making Tests A/B, digit span forward/backward, Boston Naming Test
- **Physical measurements (2):** Weight, BMI
- **Medical history (4):** Alcohol use, smoking history, cardiovascular disease, neurological conditions

Missing values were imputed using median imputation and all features were z-score normalized. Importantly, cognitive screening scores (MMSE, CDR) were excluded.

E. Model Architecture

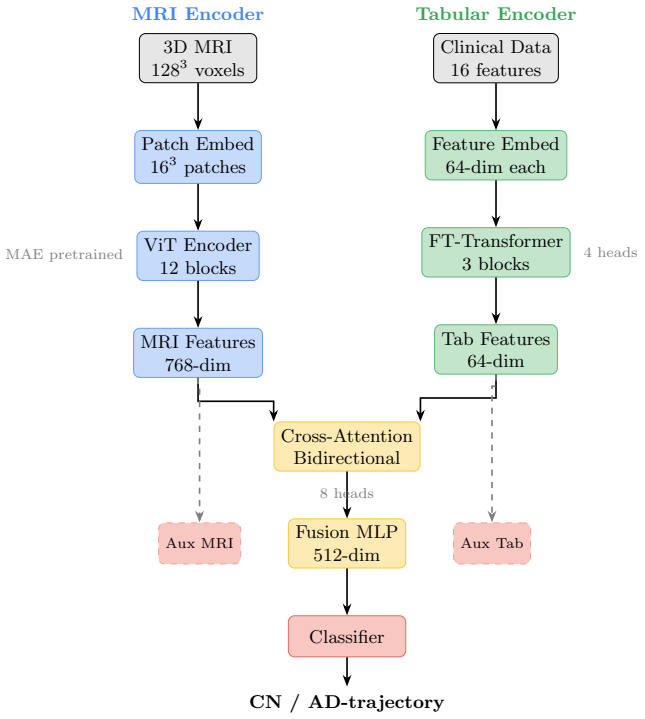

Fig. 2. Multimodal fusion architecture combining Vision Transformer (ViT) for 3D MRI and FT-Transformer for clinical features with bidirectional cross-attention fusion.

Our multimodal fusion architecture (Fig. 2) consists of three components:

MRI Encoder: We use a Vision Transformer (ViT-Base) adapted for 3D volumes, initialized with MAE-pretrained weights from Kunanbayev et al. [10]. Input volumes are divided into 16×16×16 patches, each embedded into 768-dimensional tokens. The encoder consists of 12 transformer blocks with 12 attention heads and 768-dimensional embeddings.

Tabular Encoder: We employ the FT-Transformer [9], which embeds each feature independently before processing through transformer layers. We use 3 transformer blocks with 64-dimensional embeddings and 4 attention heads, producing a 64-dimensional output.

Cross-Modal Fusion: Both modality representations are fused using sequential bidirectional cross-attention with 8 heads and hidden dimension 512. MRI features first attend to tabular features, then tabular features attend to the enriched MRI representation:

$$\hat{f}_{mri} = \text{LN}(f_{mri} + \text{CrossAttn}(f_{mri}, f_{tab}, f_{tab})) \quad (1)$$

$$f'_{mri} = \text{LN}(\hat{f}_{mri} + \text{FFN}(\hat{f}_{mri})) \quad (2)$$

$$\hat{f}_{tab} = \text{LN}(f_{tab} + \text{CrossAttn}(f_{tab}, f'_{mri}, f'_{mri})) \quad (3)$$

$$f'_{tab} = \text{LN}(\hat{f}_{tab} + \text{FFN}(\hat{f}_{tab})) \quad (4)$$

$$f_{fused} = \text{MLP}([f'_{mri}; f'_{tab}]) \quad (5)$$

where LN denotes layer normalization and semicolon denotes concatenation. Auxiliary classifiers on each branch (weighted 0.3) encourage modality-specific learning.

F. Training Details

Models were trained for up to 100 epochs with early stopping (patience of 20 epochs after a minimum of 40 epochs), saving the best checkpoint based on validation accuracy.

Optimization:

- AdamW optimizer (weight decay = 0.01)
- Learning rate: 2×10^{-5} with cosine annealing to 1×10^{-7}
- Linear warmup over 5 epochs
- Layer-wise learning rate decay factor of 0.75 for ViT fine-tuning
- Gradient clipping at norm 1.0

Regularization:

- Weighted cross-entropy loss with inverse-frequency class weights and label smoothing (ϵ = 0.1)
- Auxiliary losses on modality-specific branches (weight = 0.3)
- Test-time augmentation with 8 geometric transformations (original volume plus 7 combinations of axial/sagittal/coronal flips)

Implementation:

- Batch size: 4 (limited by 3D volume memory requirements; no gradient accumulation)
- PyTorch 2.0 on NVIDIA GPU

All experiments used 5-fold cross-validation repeated with 3 random seeds (42, 123, 456), yielding 15 training runs per configuration to assess stability.

G. Evaluation Metrics

We report accuracy, sensitivity, specificity, and AUC. Results are reported as mean $\pm$ standard deviation across 15 runs (5 folds $\times$ 3 seeds).

IV. Results

A. Classification Performance

All models were trained on the **CN vs AD-trajectory** task. We evaluate on two test sets: (1) an in-domain AD-trajectory test set matching the training distribution, and (2) a cross-task **CN vs established AD** test set containing only established AD patients without MCI converters.

Table II presents our main results. On the in-domain AD-trajectory task, our multimodal model achieved 92.4% accuracy (sensitivity: 78.8%, specificity: 96.2%) with AUC of 0.959. The high specificity indicates strong performance in correctly identifying cognitively normal subjects, while the moderate sensitivity reflects the challenge of detecting early-stage disease presentations.

The same trajectory-trained model achieved **93.3% accuracy** (sensitivity: 79.0%, specificity: 96.2%, AUC: 0.956) on the established AD test set. Notably, trajectory-based training achieves **superior sensitivity** (79.0% vs 69.4%) compared to training exclusively on established AD cases, while maintaining comparable specificity (96.2% vs 97.4%). Training on early-stage MCI converters teaches the model to detect subtle atrophy patterns, improving sensitivity on established AD cases.

TABLE II
CLASSIFICATION RESULTS: CROSS-TASK GENERALIZATION

Training Task	Test Set	Acc	Sens	Spec	AUC
Trained on CN vs AD-trajectory:					
AD-traj	AD-traj	92.4±1.0	78.8±5.5	96.2±1.1	.959±.01
AD-traj	Estab. AD	**93.3±0.9**	**79.0±5.9**	**96.2±1.1**	.956±.01
Trained on CN vs Established AD:					
Estab. AD	Estab. AD	92.9±0.7	69.4±5.4	97.4±0.9	.948±.01

Values in %.

B. Ablation Study

To quantify the contribution of each modality, we evaluated single-modality baselines using identical training protocols on the AD-trajectory task (Table III). All models used 5-fold cross-validation with 3 random seeds. The MRI-only model (ViT with MAE pretraining) achieved 83.3% accuracy (sensitivity: 56.2%, specificity: 90.8%) with AUC of 0.826. The tabular-only model (FT-Transformer) achieved 84.9% accuracy (sensitivity: 83.5%, specificity: 85.2%) with AUC of 0.925.

The multimodal fusion achieves substantial gains over both unimodal baselines: **+9.1%** accuracy over MRI-only and **+7.5%** over tabular-only. Critically, the multimodal approach achieves **96.2% specificity**—substantially higher than either modality alone—demonstrating that cross-attention fusion learns to leverage complementary strengths. The MRI-only model exhibits low sensitivity (56.2%), suggesting that structural imaging alone struggles with early-stage detection. Conversely, tabular features achieve balanced sensitivity/specificity (83.5%/85.2%), confirming that cognitive assessments and clinical variables carry significant diagnostic value.

TABLE III
ABLATION STUDY: MODALITY CONTRIBUTIONS

Model	Acc	Sens	Spec	AUC
MRI only (ViT)	83.3±0.7	56.2±1.4	90.8±0.6	.826
Tabular only (FT-T)	84.9±0.4	83.5±0.2	85.2±0.5	.925
Multimodal (Ours)	**92.4±1.0**	**78.8±5.5**	**96.2±1.1**	**.959**

AD-trajectory task, 5-fold CV. Values in %.

TABLE V
COMPARISON WITH STATE-OF-THE-ART METHODS

Study	Data	Coh	N	Acc	Sens	Spec	AUC
Ours	MRI+Tab	3	6,065	**93.3**	**79.0**	**96.2**	**.96**
Qiu [11]	MRI+Tab	8	8,916	—*	—	—	.94–.97*
VECNN [7]	MRI	1	818	94.1	95.7	97.1	—
Wen [5]	MRI	3	1k	89†	—	—	—

Acc/Sens/Spec in %. †Balanced accuracy; *Binary subtasks.

C. Cross-Cohort Generalization

To demonstrate the importance of multi-cohort training, we conducted an additional experiment training exclusively on ADNI data (903 subjects) and evaluating on held-out ADNI subjects (in-domain) as well as OASIS (1,030 subjects) and NACC (4,132 subjects) as external validation sets (Table IV).

TABLE IV
SINGLE-COHORT VS MULTI-COHORT TRAINING

Training	Test Set	Acc	Sens	Spec	AUC
ADNI-only (N=903):					
ADNI	ADNI	81.4±3.5	82.7±5.0	79.9±6.4	.90±.02
ADNI	OASIS	56.5±9.2	63.7±13.7	53.8±17.2	.64±.05
ADNI	NACC	38.5±13.9	98.0±2.3	29.4±16.4	.89±.05
Multi-cohort (N=6,065):					
Multi	All	**92.4±1.0**	**78.8±5.5**	**96.2±1.1**	**.959±.01**

All results: 5-fold CV × 3 seeds. Values in %.

The results reveal substantial cross-cohort generalization gaps for single-cohort models: while ADNI-only training achieves 81.4±3.5% accuracy in-domain, performance drops to 56.5±9.2% on OASIS (-24.9 points) and 38.5±13.9% on NACC (-42.9 points). The high variance on external cohorts (SD up to 13.9%) reflects inconsistent generalization across folds. NACC exhibits notable miscalibration despite high AUC (0.89): the model achieves 98% sensitivity but only 29% specificity, incorrectly classifying 70% of healthy controls as diseased. This arises from class distribution shift (ADNI: 47% CN; NACC: 87% CN), causing decision threshold mismatch. These results demonstrate that single-cohort models overfit to scanner-specific and population-specific patterns. By comparison, our multi-cohort approach achieves **92.4% accuracy**—surpassing even ADNI in-domain performance by 11 points—by learning cohort-invariant features from diverse acquisition protocols and demographics, confirming that multi-cohort training is essential for clinically deployable AD classification systems.

D. Comparison with Literature

Table V compares our results with prior work using proper subject-level splitting.

We achieve 93.3% accuracy on established AD detection and 92.4% on the AD-trajectory task. Most prior work evaluates on single cohorts with <1,000 subjects. While approaches like VECNN [7] achieve strong single-cohort performance (94.1% accuracy on ADNI), our external validation experiments (Table IV) reveal the critical importance of multi-cohort training. Qiu et al. [11] achieved similar AUC performance (0.94–0.97) using separate CNN and CatBoost models, whereas our approach employs end-to-end learning with cross-attention fusion for joint MRI-clinical representation learning.

V. DISCUSSION

A. Cross-Task Generalization

Our results reveal that training on the heterogeneous AD-trajectory task (including MCI converters) yields superior performance for established AD detection compared to training exclusively on established AD cases. Models trained on AD-trajectory achieve 0.956 AUC on established AD versus 0.948 for models trained directly on established AD (Table II), with significantly higher sensitivity (79.0% vs 69.4%) despite slightly lower specificity (96.2% vs 97.4%).

This demonstrates that training on challenging early-stage cases forces the model to learn discriminative features that generalize to pronounced disease. MCI converters exhibit subtle atrophy patterns near the noise floor of normal aging variability, requiring fine-grained feature extraction. When applied to established AD cases with pronounced atrophy, these learned features transfer effectively without relying on gross anatomical changes. The trajectory-trained model achieves 93.3% accuracy on established AD detection, only 0.9 points below its in-domain performance (92.4%), supporting the clinical utility of heterogeneous training for robust early detection.

B. Limitations

Several limitations should be acknowledged:

Class imbalance: The 78%/22% CN/AD ratio affects sensitivity. Future work could explore oversampling or focal loss.

Missing data: Some tabular features have incomplete coverage across cohorts (23-100%), handled via median imputation which may lose information.

Interpretability: Future work could employ feature attribution methods (e.g., SHAP values) to quantify individual clinical feature contributions and identify the most predictive variables.

External validation: While our cross-cohort experiment (Table IV) demonstrates the importance of multi-cohort training, all three cohorts participate in cross-validation. A fully independent held-out cohort used only for testing would provide stronger generalization evidence.

VI. Conclusion

We presented a multimodal deep learning framework for Alzheimer's disease classification combining Vision Transformer and Feature Tokenizer Transformer with cross-attention fusion. Validated on 6,065 subjects from three independent cohorts with strict subject-level splitting, our approach achieves 92.4% accuracy on the challenging AD-trajectory task and 93.3% on established AD detection. Multimodal fusion provides substantial gains over unimodal baselines (+9.1% over MRI-only, +7.5% over tabular-only). Training on heterogeneous AD-trajectory cases yields improved generalization to established AD (AUC: 0.956 vs 0.948), demonstrating that early-stage training enhances late-stage detection. External validation experiments confirm the necessity of multi-cohort training for robust clinical deployment, with single-cohort models exhibiting performance drops of 25-43 points on held-out cohorts.

Acknowledgment

*Data used in preparation of this article were obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu). As such, the investigators within the ADNI contributed to the design and implementation of ADNI and/or provided data but did not participate in the analysis or writing of this report. A complete listing of ADNI investigators can be found at: http://adni.loni.usc.edu/wp-content/uploads/how_to_apply/ADNI_Acknowledgement_List.pdf

Data collection and sharing for the Alzheimer's Disease Neuroimaging Initiative (ADNI) is funded by the National Institute on Aging (National Institutes of Health Grant U19AG024904). The grantee organization is the Northern California Institute for Research and Education. In the past, ADNI has also received funding from the National Institute of Biomedical Imaging and Bioengineering, the Canadian Institutes of Health Research, and private sector contributions through the Foundation for the National Institutes of Health (FNIH) including generous contributions from the following: AbbVie, Alzheimer's Association; Alzheimer's Drug Discovery Foundation; Araclon Biotech; BioClinica, Inc.; Biogen; Bristol-Myers Squibb Company; CereSpir, Inc.; Cogstate; Eisai Inc.; Elan Pharmaceuticals, Inc.; Eli Lilly and Company; EuroImmun; F. Hoffmann-La Roche Ltd and its affiliated company Genentech, Inc.; Fujirebio; GE Healthcare; IXICO Ltd.; Janssen Alzheimer Immunotherapy Research & Development, LLC.; Johnson & Johnson Pharmaceutical Research & Development LLC.; Lumosity; Lundbeck; Merck & Co., Inc.; Meso Scale Diagnostics, LLC.; NeuroRx Research; Neurotrack Technologies; Novartis Pharmaceuticals Corporation; Pfizer Inc.; Piramal Imaging; Servier; Takeda Pharmaceutical Company; and Transition Therapeutics.

Data were provided in part by OASIS-3: Longitudinal Multimodal Neuroimaging: Principal Investigators: T. Benzinger, D. Marcus, J. Morris; NIH P30 AG066444, P50 AG00561, P30 NS09857781, P01 AG026276, P01 AG003991, R01 AG043434, UL1 TR000448, R01 EB009352, P30 AG066444, AW00006993.

The NACC database is funded by NIA/NIH Grant U24 AG072122. NACC data are contributed by the NIA-funded ADRCs: P30 AG062429 (PI James Brewer, MD, PhD), P30 AG066468 (PI Oscar Lopez, MD), P30 AG066421 (PI Bradley Hyman, MD, PhD), P30 AG066509 (PI Thomas Grabowski, MD), P30 AG066514 (PI Mary Sano, PhD), P30 AG066530 (PI Helena Chui, MD), P30 AG066507 (PI Marilyn Albert, PhD), P30 AG066444 (PI John Morris, MD), P30 AG066518 (PI Jeffrey Kaye, MD), P30 AG066512 (PI Thomas Wisniewski, MD), P30 AG066462 (PI Scott Small, MD), P30 AG072979 (PI David Wolk, MD), P30 AG072972 (PI Charles DeCarli, MD), P30 AG072976 (PI Andrew Saykin, PsyD), P30 AG072975 (PI David Bennett, MD), P30 AG072978 (PI Neil Kowall, MD), P30 AG072977 (PI Robert Vassar, PhD), P30 AG066519 (PI Frank LaFerla, PhD), P30 AG062677 (PI Ronald Petersen, MD, PhD), P30 AG079280 (PI Eric Reiman, MD), P30 AG062422 (PI Gil Rabinovici, MD), P30 AG066511 (PI Allan Levey, MD, PhD), P30 AG072946 (PI Linda Van Eldik, PhD), P30 AG062715 (PI Sanjay Asthana, MD, FRCP), P30 AG072973 (PI Russell Swerdlow, MD), P30 AG066506 (PI Todd Golde, MD, PhD), P30 AG066508 (PI Stephen Strittmatter, MD, PhD), P30 AG066515 (PI Victor Henderson, MD, MS), P30 AG072947 (PI Suzanne Craft, PhD), P30 AG072931 (PI Henry Paulson, MD, PhD), P30 AG066546 (PI Sudha Seshadri, MD), P20 AG068024 (PI Erik Roberson, MD, PhD), P20 AG068053 (PI Justin Miller, PhD), P20 AG068077 (PI Gary Rosenberg, MD), P20 AG068082 (PI Angela Jefferson, PhD), P30 AG072958 (PI Heather Whitson, MD), P30 AG072959 (PI James Leverenz, MD). SCAN is a multi-institutional project that was funded as a U24 grant (AG067418) by the National Institute on Aging in May 2020. Data collected by SCAN and shared by NACC are contributed by the NIA-funded ADRCs as follows: Arizona Alzheimer's Center - P30 AG072980 (PI: Eric Reiman, MD); R01 AG069453 (PI: Eric Reiman (contact), MD); P30 AG019610 (PI: Eric Reiman, MD); and the State of Arizona which provided additional funding supporting our center; Boston University - P30 AG013846 (PI Neil Kowall MD); Cleveland ADRC - P30 AG062428 (James Leverenz, MD); Cleveland Clinic, Las Vegas - P20AG068053; Columbia - P50 AG008702 (PI Scott Small MD); Duke/UNC ADRC - P30 AG072958; Emory University - P30AG066511 (PI Levey Allan, MD, PhD); Indiana University - R01 AG19771 (PI Andrew Saykin, PsyD); P30 AG10133 (PI Andrew Saykin, PsyD); P30 AG072976 (PI Andrew Saykin, PsyD); R01 AG061788 (PI Shannon Risacher, PhD); R01 AG053993 (PI Yu-Chien Wu, MD, PhD); U01 AG057195 (PI Liana Apostolova, MD); U19 AG063911 (PI Bradley Boeve, MD); and the Indiana University Department of Radiology and Imaging Sciences; Johns Hopkins - P30 AG066507 (PI Marilyn Albert, Phd.); Mayo Clinic - P50 AG016574 (PI Ronald Petersen MD PhD); Mount Sinai - P30 AG066514 (PI Mary Sano, PhD); R01 AG054110 (PI Trey Hedden, PhD); R01 AG053509 (PI Trey Hedden, PhD); New York University - P30AG066512-01S2 (PI Thomas Wisniewski, MD); R01AG056031 (PI Ricardo Osorio, MD); R01AG056531 (PIs Ricardo Osorio, MD;

Girardin Jean-Louis, PhD); Northwestern University - P30 AG013854 (PI Robert Vassar PhD); R01 AG045571 (PI Emily Rogalski, PhD); R56 AG045571, (PI Emily Rogalski, PhD); R01 AG067781, (PI Emily Rogalski, PhD); U19 AG073153, (PI Emily Rogalski, PhD); R01 DC008552, (M.-Marsel Mesulam, MD); R01 AG077444, (PIs M.-Marsel Mesulam, MD, Emily Rogalski, PhD); R01 NS075075 (PI Emily Rogalski, PhD); R01 AG056258 (PI Emily Rogalski, PhD); Oregon Health and Science University - P30 AG008017 (PI Jeffrey Kaye MD); R56 AG074321 (PI Jeffrey Kaye, MD); Rush University - P30 AG010161 (PI David Bennett MD); Stanford - P30AG066515; P50 AG047366 (PI Victor Henderson MD MS); University of Alabama, Birmingham - P20; University of California, Davis - P30 AG10129 (PI Charles DeCarli, MD); P30 AG072972 (PI Charles DeCarli, MD); University of California, Irvine - P50 AG016573 (PI Frank LaFerla PhD); University of California, San Diego - P30AG062429 (PI James Brewer, MD, PhD); University of California, San Francisco - P30 AG062422 (Rabinovici, Gil D., MD); University of Kansas - P30 AG035982 (Russell Swerdlow, MD); University of Kentucky - P30 AG028283-15S1 (PIs Linda Van Eldik, PhD and Brian Gold, PhD); University of Michigan ADRC - P30AG053760 (PI Henry Paulson, MD, PhD) P30AG072931 (PI Henry Paulson, MD, PhD) Cure Alzheimer's Fund 200775 - (PI Henry Paulson, MD, PhD) U19 NS120384 (PI Charles DeCarli, MD, University of Michigan Site PI Henry Paulson, MD, PhD) R01 AG068338 (MPI Bruno Giordani, PhD, Carol Persad, PhD, Yi Murphey, PhD) S10OD026738-01 (PI Douglas Noll, PhD) R01 AG058724 (PI Benjamin Hampstead, PhD) R35 AG072262 (PI Benjamin Hampstead, PhD) W81XWH2110743 (PI Benjamin Hampstead, PhD) R01 AG073235 (PI Nancy Chiaravalloti, University of Michigan Site PI Benjamin Hampstead, PhD) 1I01RX001534 (PI Benjamin Hampstead, PhD) IRX001381 (PI Benjamin Hampstead, PhD); University of New Mexico - P20 AG068077 (Gary Rosenberg, MD); University of Pennsylvania - State of PA project 2019NF4100087335 (PI David Wolk, MD); Rooney Family Research Fund (PI David Wolk, MD); R01 AG055005 (PI David Wolk, MD); University of Pittsburgh - P50 AG005133 (PI Oscar Lopez MD); University of Southern California - P50 AG005142 (PI Helena Chui MD); University of Washington - P50 AG005136 (PI Thomas Grabowski MD); University of Wisconsin - P50 AG033514 (PI Sanjay Asthana MD FRCP); Vanderbilt University - P20 AG068082; Wake Forest - P30AG072947 (PI Suzanne Craft, PhD); Washington University, St. Louis - P01 AG03991 (PI John Morris MD); P01 AG026276 (PI John Morris MD); P20 MH071616 (PI Dan Marcus); P30 AG066444 (PI John Morris MD); P30 NS098577 (PI Dan Marcus); R01 AG021910 (PI Randy Buckner); R01 AG043434 (PI Catherine Roe); R01 EB009352 (PI Dan Marcus); UL1 TR000448 (PI Brad Evanoff); U24 RR021382 (PI Bruce Rosen); Avid Radiopharmaceuticals / Eli Lilly; Yale - P50 AG047270 (PI Stephen Strittmatter MD PhD); R01AG052560 (MPI: Christopher van Dyck, MD; Richard Carson, PhD); R01AG062276 (PI: Christopher van Dyck, MD); 1Florida - P30AG066506-03 (PI Glenn Smith, PhD); P50 AG047266 (PI Todd Golde MD PhD).

REFERENCES

[1] World Health Organization, "Dementia," 2023. [Online]. Available: https://www.who.int/health-topics/dementia

[2] V. M. Young, S. Gates, L. Y. Garcia, and A. Salardini, "Data leakage in deep learning for Alzheimer's disease diagnosis: A scoping review of methodological rigor and performance inflation," *Diagnostics*, vol. 15, no. 18, p. 2348, 2025.

[3] E. Yagis, A. G. S. De Herrera, and L. Citi, "Effect of data leakage in brain MRI classification using 2D convolutional neural networks," *Sci. Rep.*, vol. 11, no. 22544, 2021.

[4] S. Qiu et al., "Development and validation of an interpretable deep learning framework for Alzheimer's disease classification," *Brain*, vol. 143, no. 6, pp. 1920–1933, 2020.

[5] J. Wen et al., "Convolutional neural networks for classification of Alzheimer's disease: Overview and reproducible evaluation," *Med. Image Anal.*, vol. 63, p. 101694, 2020.

[6] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," in *Proc. ICLR*, 2021.

[7] Z. Zhao et al., "Vision transformer-equipped convolutional neural networks for automated Alzheimer's disease diagnosis using 3D MRI scans," *Front. Neurol.*, vol. 15, p. 1490829, 2024.

[8] K. He, X. Chen, S. Xie, Y. Li, P. Dollar, and R. Girshick, "Masked autoencoders are scalable vision learners," in *Proc. CVPR*, 2022.

[9] Y. Gorishniy, I. Rubachev, V. Khrulkov, and A. Babenko, "Revisiting deep learning models for tabular data," in *Proc. NeurIPS*, 2021.

[10] K. Kunanbayev, V. Shen, and D. S. Kim, "Training ViT with limited data for Alzheimer's disease classification: An empirical study," in *Proc. MICCAI*, 2024, pp. 339–349. doi: 10.1007/978-3-031-72390-2_32.

[11] S. Qiu et al., "Multimodal deep learning for Alzheimer's disease dementia assessment," *Nat. Commun.*, vol. 13, p. 3404, 2022.

[12] S. G. Mueller et al., "The Alzheimer's disease neuroimaging initiative," *Neuroimaging Clin. N. Am.*, vol. 15, no. 4, pp. 869–877, 2005.

[13] P. J. LaMontagne et al., "OASIS-3: Longitudinal neuroimaging, clinical, and cognitive dataset for normal aging and Alzheimer disease," *medRxiv*, 2019. doi: 10.1101/2019.12.13.19014902.

[14] D. L. Beekly et al., "The National Alzheimer's Coordinating Center (NACC) database: the Uniform Data Set," *Alzheimer Dis. Assoc. Disord.*, vol. 21, no. 3, pp. 249–258, 2007.

[15] A. Hoopes, J. S. Mora, A. V. Dalca, B. Fischl, and M. Hoffmann, "SynthStrip: Skull-stripping for any brain image," *NeuroImage*, vol. 260, p. 119474, 2022.

[16] C. R. Jack Jr. et al., "NIA-AA Research Framework: Toward a biological definition of Alzheimer's disease," *Alzheimer's Dement.*, vol. 14, no. 4, pp. 535–562, 2018.

[17] C. R. Jack Jr., R. C. Petersen, Y. Xu, et al., "Rates of hippocampal atrophy correlate with change in clinical status in aging and AD," *Neurology*, vol. 55, no. 4, pp. 484–489, 2000.

[18] C. Misra, Y. Fan, and C. Davatzikos, "Baseline and longitudinal patterns of brain atrophy in MCI patients, and their use in prediction of short-term conversion to AD: Results from ADNI," *NeuroImage*, vol. 44, no. 4, pp. 1415–1422, 2009.